# High Validity, Low Transfer: Anchored Evaluation of Diffusion Counterfactual Explanations for Chest Radiographs

## Supplementary Material

## R. Reproducibility Appendix

Every value in this appendix is read from the code and configuration of the default-configuration runs of the main paper (2-model guidance ensemble) and checked against their run logs, recorded configuration and checkpoints. Settings that differ between code versions are listed in Sec. R5 with the value the default runs used.

### R1. Reward

Notation: $x$ is the episode's original image (fixed for the episode), $x_{t+1}$ the image after policy step $t$, $d_t = |f(x_t) - c|$ with $f$ the guidance classifier, $S$ the soft lung mask of $x$ (computed once per episode, Table R4), and $p_O(\cdot)$ the softmax, clamped to $[10^{-8}, 1]$, of the frozen YOLOv8n feature vector (Table R4). Means and sums run over the $256 \times 256$ pixels. The per-step reward is

$$r_t = k_c \sum_j \lambda_j R_j,$$

with $k_c = 2.25$ for n→p ($c = 1$) and $k_c = 1$ for p→n ($c = 0$), applied to every term.

$\lambda_{\text{sp}}(u) = 0$ outside the sparsity arm. In that arm it is 0 for updates $u < \text{round}(0.2U)$, rises linearly to 2.0 at $u = \text{round}(0.6U)$ and stays there ($U = 100$ updates). A safeguard measures CV every 10 updates, from 10 updates before the ramp until its end, on 24 held-out far-pool images with the deterministic policy and a budget of $\min(4, T)$ steps. If CV falls more than 0.08 below its pre-ramp value, it freezes $\lambda_{\text{sp}}$ for the rest of training at the value of the last probe that passed. In the trained sparsity arm the safeguard froze $\lambda_{\text{sp}}$ at 0 for n→p at update 20, before the ramp moved; p→n ramped to 2.0.

### R2. Action mapping

The actor outputs the mean $\mu$ of a 9-dimensional diagonal Gaussian over a raw, unbounded vector $a \in \mathbb{R}^9$. During training $a$ is sampled; at evaluation $a = \mu$. Each control is a fixed transform of $a$, with $\sigma$ the logistic sigmoid (Table R2). The raw vector is never clipped. Ranges are open intervals, because $\sigma$ never reaches 0 or 1.

*Discrete controls.* No Bernoulli distribution is used. $a_4$ and $a_5$ are Gaussian like the other dimensions and are thresholded deterministically, $e_i = \mathbf{1}[\sigma(a) > 0.5] = \mathbf{1}[a > 0]$, identically in training and evaluation. The PPO log-probability is that of the raw Gaussian sample, with no correction for the sigmoid or the threshold. Stochasticity in $e_1, e_2$ comes only from Gaussian exploration during training.

*Per-pass schedule.* Each policy step runs $n = 2 + e_1 + e_2$ chained generator passes, each taking the previous pass's output as its input. Pass $k = 0, \ldots, n-1$ uses

$$s_k = \min\big(0.9,\ s\,\beta\,(1 + 0.15k)\big),$$
$$g_{c,k} = g_c\, b\,(1 + 0.35k),$$
$$g_{p,k} = g_p\ \max(1/b,\ 0.6)\ \max(1 - 0.15k,\ 0.4),$$

and the same ramp start $r$. The applied strength $s_k$ therefore ranges over $(0.17, 0.9]$ rather than Table 1's $(0.20, 0.70)$. Neither floor in $g_{p,k}$ binds: $1/b > 0.769$ and $1 - 0.15k \geq 0.55$ for $k \leq 3$.

### R3. PPO

### R4. Generator, LoRA, and background lock

### R5. Settings that differ between code versions

Table R5 lists every setting above that differs between code versions, with the value used by the default runs. The 3-model, sparsity and Grad-CAM runs came from uncommitted code states, identified by their configuration hash and logged flags.

Also recorded for the default runs: the patient-level split (80% of patients for training and 20% for test, seed 0, persisted with a dataset fingerprint; one image per patient, 21,347 training and 5,337 test images, unstratified); the random seeds (Table R3); the LHR implementation and the per-shard configuration stamp that fixes the transfer-ratio denominator (Sec. R6); and a run manifest for every training and evaluation run, giving its code revision and configuration. The random-action and no-policy arms compared with the single-classifier policy are evaluation-only and reuse that run's generator, `rsna` guidance and evaluation groups. The no-policy, fixed-schedule and second-seed runs compared with the default (main-paper Sec. 5.2, 5.4; Supp. E5) instead use the default's guidance, evaluation groups, images and per-image seeds, with each direction's own adapter (the second seed's two directions share one adapter). The held-out transfer point estimates were recomputed from the saved evaluation shards and agree with the originally logged values to within $10^{-4}$; the differences come from GPU type and affect only the last printed digit. The bootstrap intervals reported in the main paper come from this recomputation (Sec. R6).

### R6. Metric definitions

Notation: $X_n = \{x : f(x) < 0.2\}$ and $X_p = \{x : f(x) > 0.8\}$ are the first $N \leq 200$ images of each kind, in dataset

Table R1. Reward terms as implemented. $B$ is an $L_1$ edit budget on validity; with $B = \infty$ validity depends on $d_{t+1}$ alone.

| Term | Implemented formula | Weight $\lambda_j$ |
|---|---|---|
| $R_{\text{prog}}$ (progress) | $d_t - d_{t+1}$ | 1.5 |
| $R_{\text{val}}$ (distance) | $-d_{t+1}$ | 2.0 |
| $R_{\text{close2}}$ (squared distance) | $-d_{t+1}^2$ | 2.5 |
| $R_{\text{fid}}$ (fidelity) | $-\operatorname{mean}_p \lvert x_{t+1} - x\rvert$ (mean pixel $L_1$) | 2.0 |
| $R_{\text{pres}}$ (preservation) | $-\Big(\frac{\sum_p S\,\lvert x_{t+1} - x\rvert}{\sum_p S + 10^{-8}} + \eta\,\mathrm{KL}\big(p_O(x)\,\|\,p_O(x_{t+1})\big)\Big)$, $\eta = 0.5$; KL summed over feature dimensions | 3.0 |
| $R_{\text{bud}}$ (per-step cost) | $-1$ | 0.1 |
| $R_{\text{hit}}$ (validity bonus) | $\mathbf{1}\big[d_{t+1} < \epsilon_{\text{stop}} \wedge \operatorname{mean}_p \lvert x_{t+1} - x\rvert < B\big]$, $\epsilon_{\text{stop}} = 0.08$, $B = \infty$; validity also ends the episode, so this is paid at most once, on the last step | 2.0 |
| $R_{\text{stop}}$ (stop bonus) | $R_{\text{hit}} \cdot \mathbf{1}[\tau_t > 0.5]$, $\tau_t = \sigma(a_3)$ | 2.0 |
| $R_{\text{sp}}$ (sparsity arm only) | $-\operatorname{mean}_p \lvert x_{t+1} - x\rvert$; $(1 - \text{CAM})$-weighted and normalized by the weight sum when a Grad-CAM mask is active | $\lambda_{\text{sp}}(u)$ |

Table R2. Raw-action-to-control mapping, checked cell by cell against Table 1 of the main paper.

| Dim. | Control | Mapping | Range | Main-paper Table 1 |
|---|---|---|---|---|
| $a_0$ | DDIM strength $s$ (base value) | $0.20 + 0.50\,\sigma(a_0)$ | $(0.20, 0.70)$ | matches |
| $a_1$ | classifier-guidance scale $g_c$ | $0.30 + 0.90\,\sigma(a_1)$ | $(0.30, 1.20)$ | matches |
| $a_2$ | preservation-guidance scale $g_p$ | $0.30 + 1.20\,\sigma(a_2)$ | $(0.30, 1.50)$ | matches |
| $a_3$ | stop signal $\tau$ | $\sigma(a_3)$ | $(0, 1)$ | matches |
| $a_4, a_5$ | extra passes $e_1, e_2$ | $\mathbf{1}[\sigma(a) > 0.5]$ | $\{0, 1\}$ | matches |
| $a_6$ | strength boost $\beta$ | $0.85 + 0.30\,\sigma(a_6)$ | $(0.85, 1.15)$ | matches |
| $a_7$ | guidance bias $b$ | $0.70 + 0.60\,\sigma(a_7)$ | $(0.70, 1.30)$ | matches |
| $a_8$ | ramp start fraction $r$ | $0.10 + 0.40\,\sigma(a_8)$ | $(0.10, 0.50)$ | matches |

order, selected by the guidance classifier $f$ rather than by label. $f$ is the ensemble's mean probability in ensemble arms. n→p edits $X_n$ and p→n edits $X_p$, with $s = +1$ and $s = -1$ respectively. Images are grayscale, $256 \times 256$, in $[0, 1]$, and $\delta = |x_c - x|$ per pixel.

**Counterfactual Validity.**

$$\mathrm{CV} = \frac{1}{N}\sum_i \mathbf{1}\big[s\,(f(x_c^i) - f(x^i)) \geq \tau \wedge B_i\big], \quad \tau = 0.8,$$

with $B_i$: $f(x^i) \leq 0.5 < f(x_c^i)$ for n→p and $f(x^i) \geq 0.5 > f(x_c^i)$ for p→n. Pairs with a non-finite score are dropped. Mean CV is the unweighted mean of the two directions. On $X_n$ and $X_p$ the margin condition already implies $B_i$: with $f \in [0, 1]$, $s\,(f(x_c^i) - f(x^i)) \geq 0.8$ forces $f(x_c) \geq 0.8$ for n→p and $f(x_c) \leq 0.2$ for p→n, and the source side holds by construction of the groups. The tightening relative to a direction-agnostic CV therefore comes from the sign requirement alone.

**Fraction of pixels changed.** $\phi_i = \frac{1}{HW}\sum_p \mathbf{1}[\delta_p > 0.01]$, averaged over pairs. Pairs with $\sum_p \delta_p = 0$ are skipped, both here and for edit concentration.

**Edit concentration.** $C_i = \sum_{p \in \mathrm{top}_k(\delta)} \delta_p \big/ \sum_p \delta_p$ with $k = \mathrm{round}(0.05 \cdot 65{,}536) = 3{,}277$, averaged over pairs. Spatially uniform change scores exactly 0.05. As an illustrative heuristic only, not a threshold, a compact lesion-shaped edit would be expected to put roughly half or more of its mass in the top 5% of pixels. We do not use 0.5 as a decision boundary.

**Identity preservation.** Both scores are computed per pair (source image vs. its counterfactual) and averaged. *MS-SSIM*: five scales with $2 \times 2$ average pooling between them; an $11 \times 11$ Gaussian window, $\sigma = 1.5$, zero padding; $C_1 = 0.01^2$ and $C_2 = 0.03^2$ (data range 1); scale weights $(0.0448, 0.2856, 0.3001, 0.2363, 0.1333)$. The result is $\prod_{j=1}^{4} \mathrm{cs}_j^{w_j} \cdot \mathrm{ssim}_5^{w_5}$, with each factor clamped at $\geq 10^{-6}$. *HF-corr*: $h(x) = x - G_{11,1.5} * x$, $r$ the Pearson correlation of $h(x)$ and $h(x_c)$ over all pixels, reported as $(1 + r)/2$.

**FID.** $\mathrm{FID}(A, B) = \|\mu_A - \mu_B\|^2 + \mathrm{Tr}\big(\Sigma_A + \Sigma_B - 2(\Sigma_A\Sigma_B)^{1/2}\big)$. Here $\Sigma$ is the unbiased sample covariance plus $10^{-6}I$, and the real part of the matrix square root is taken. Features are the global-average-pooled output

Table R3. PPO settings. One policy per direction, trained independently with identical settings.

| Setting | Value |
|---|---|
| Episodes per direction | 800 |
| Episodes per update | 8 (100 updates) |
| Minibatch | full batch: one gradient step per epoch on all transitions of the update ($\leq 8T$); no minibatching |
| PPO epochs per update | 4 |
| Optimizer | AdamW, learning rate $3 \times 10^{-4}$, constant; other arguments at PyTorch defaults |
| Clip $\epsilon$ | 0.2 |
| GAE | $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$; no bootstrap past episode end; advantages normalized to zero mean and unit variance over the update |
| Value loss | weight 0.5; MSE to the shaped target $r_t + 0.2\,A_t$ if $\tau_t \leq 0.5$, else $r_t$, where $A_t$ is the unnormalized GAE advantage (not the GAE return) |
| Entropy bonus | weight 0.01; entropy of the raw 9-d Gaussian, summed over dimensions |
| Auxiliary heads | weight 0.1 each. Feasibility: sigmoid, BCE to $\mathbf{1}[\hat{R}_t > 0]$ ($\hat{R}_t$ the GAE return). Progress: linear, MSE to $r_t + \gamma V_{t+1}$ ($V_{t+1} = 0$ at episode end). Stop: sigmoid, BCE to $\mathbf{1}[\tau_t > 0.5]$ |
| Saturation penalty | $0.01 \cdot \text{mean ReLU}(\lvert\mu\rvert - 3)^2$ over raw actor means |
| Gradient clipping | global norm 0.5 |
| Policy std | state-independent learnable per-dimension $\log\sigma$, initialized to $-0.5$; $\sigma$ floored at $10^{-3}$ |
| Trunk | MLP 73,731 $\to$ 512 $\to$ 256, ReLU; 37,885,462 policy parameters in total. The alternative convolutional trunk has 197,910 |
| Heads | linear from the 256-d trunk feature: actor (9), value (1), feasibility, progress, stop (1 each) |
| Observation | $[z_x,\ z_{x_t},\ t/T,\ d_t,\ c,\ S(x_t)]$: two VAE-posterior samples ($2 \times 4{,}096$), 3 scalars, and the $256^2$ lung mask of $x_t$; 73,731 entries |
| Step budget $T$ | 25, in training and evaluation |
| Termination | $d_{t+1} < \epsilon_{\text{stop}} = 0.08$ (with $B = \infty$) or $t = T$ |
| Training images | drawn uniformly with replacement, one per episode, from training-fold images with $\lvert f(x) - c\rvert > 0.5$, excluding the first 24 of that pool, which are held out for the sparsity-arm probe in every arm |
| Final policy | weights after the last update; no checkpoint selection |
| Evaluation | mean action ($a = \mu$), same $T$ and termination; per-image seed = base seed 0 + global image index ($+10^6$ for p$\to$n) |
| Random seeds | split 0; LoRA fine-tuning 0; PPO 1 (n$\to$p) and 2 (p$\to$n), i.e. base 0 plus a per-direction offset; cuDNN deterministic mode on, full determinism off, so repeated runs are not guaranteed to be bit-identical, and re-running the identical protocol on different hardware regenerates different images, so aggregate metrics move by small amounts (two runs of the default p$\to$n protocol differ in LHR lift by $+0.016$, 95% CI $[-0.045, +0.098]$, on 148 images); evaluation base seed 0 |

(1024-d) of the `features` module of TorchXRayVision DenseNet-121 `densenet121-res224-rsna`, applied to the image rescaled to $[-1024, 1024]$ and resized bilinearly to $224^2$. The main paper reports the cross-group values $\text{FID}(X_p, X_c^{\text{n}\to\text{p}})$ and $\text{FID}(X_n, X_c^{\text{p}\to\text{n}})$, where the "real target class" is the classifier-selected group, not the label. The feature extractor is the `rsna` model, which also guides generation in every trained arm (the 3-model ensemble is `rsna`, `chex`, `mimic_nb`), so FID is not independent of guidance.

**Held-out transfer ratio.** For validator $v \in \{\texttt{nih}, \texttt{pc}, \texttt{mimic_ch}\}$ (TorchXRayVision DenseNet-121, Pneumonia output) and direction $s$,

$$\rho_v = \frac{\frac{1}{N}\sum_i s\left(f_v(x_c^i) - f_v(x^i)\right)}{\frac{1}{N}\sum_i s\left(f_g(x_c^i) - f_g(x^i)\right)},$$

over the same $N$ images. $f_g$ is the guidance classifier (mean probability over members for an ensemble), and $\rho_v$ is undefined when the denominator is exactly zero. No validator is a guidance member in any reported arm. The denominator must use the classifier(s) that actually guided generation: it is computed from the guidance roster in each evaluation shard's configuration stamp, which records the guidance flag and member list for the default and later arms; the single-classifier arm's shards predate this key and are treated as single-classifier guidance, which its training log confirms. The 3-validator mean in main-paper Tables 2 and 4 is the unweighted mean of the per-validator ratios.

Table R4. Generator settings. $x_a$ is the anchor image: the episode's original image.

| Setting | Value |
|---|---|
| Backbone | SD v1.5 (`runwayml/stable-diffusion-v1-5`) VAE and UNet, fp16, frozen except LoRA. Text encoder unused. $256^2$ grayscale input replicated to 3 channels and mapped to $[-1, 1]$; $4 \times 32 \times 32$ latent drawn from the VAE posterior (sample, not mean) |
| Conditioning MLP | $[c,$ latent average-pooled to $64] \in \mathbb{R}^{65} \to 256 \to 768$, SiLU; one cross-attention token; 214,272 parameters, fp32, trained jointly with LoRA |
| LoRA | rank 8, $\alpha = 16$, dropout 0.05, on `to_q`, `to_k`, `to_v`, `to_out.0` of the attention layers (1,594,368 LoRA parameters) |
| LoRA fine-tuning | $\epsilon$-prediction MSE only; timestep uniform over the 50 DDIM timesteps; 30 epochs, batch 8, AdamW with learning rate $10^{-4}$; class-balanced sampling with replacement; fp16 autocast with gradient scaling; gradient-norm clip 1.0. Training images: the 21,347 images of the training fold. Each direction fine-tunes its own adapter with this recipe and seed, and the resulting weights differ. As the Frobenius norm of the difference over all LoRA tensors, relative to the n→p adapter's norm: 0.385 (default), $1.18 \times 10^{-2}$ (single-classifier), $3.94 \times 10^{-3}$ (3-model), $1.49 \times 10^{-3}$ (sparsity), and 0 for the Grad-CAM arm, whose two adapters are identical; the default arm's difference is 33 to 259 times the other arms'. The default and Grad-CAM arms are evaluated with each direction's own adapter, the other arms with the n→p adapter for both directions. All default-arm numbers in the main paper come from the evaluation with each direction's own adapter. An earlier evaluation of the default with the n→p adapter for both directions, on the same images and seeds, is used only where the default is compared with the three arms evaluated that way (main-paper Table 4); against it, the per-direction evaluation has LHR lift lower by 0.077 on the 148 annotated images (paired 95% CI $[-0.159, +0.001]$) and p→n CV lower by 0.045 (19 vs. 10 discordant images, exact McNemar $p = 0.14$); its n→p CV is 0.945 vs. 0.960 (3 vs. 0 discordant, $p = 0.25$). Two runs of the identical matched-adapter protocol, partly on different GPU types, differ by $+0.016$ in LHR lift ($[-0.045, +0.098]$) and by 0.015 in p→n CV (6 vs. 9 discordant, $p = 0.61$), so the changes are not attributed to the adapter alone |
| DDIM | SD1.5 scheduler configuration, $T_{\text{diff}} = 50$ timesteps; pass $k$ keeps the last $\max(1, \lfloor 50\, s_k \rfloor)$ and starts from the pass input's latent, forward-noised to the first kept timestep. DDIM $\eta = 0$ (not set in code; library default) |
| Guidance losses | $\mathcal{L}_{\text{cls}} = \text{BCE}(f(D(\hat{z}_0)), c)$; $\mathcal{L}_{\text{pres}} = \dfrac{\sum S\,\lvert D(\hat{z}_0) - x_a\rvert}{\sum S + 10^{-8}} + \eta\,\text{MSE}\big(O(D(\hat{z}_0)), O(x_a)\big)$, $\eta = 0.5$ (the reward uses KL on softmaxed features instead, Table R1) |
| Guidance update | $z_{t-1} \leftarrow z_{t-1} - w(i)\, m \odot \big(1.15\, \alpha_{\text{cls}}\, g_{c,k} \nabla_{\hat{z}_0} \mathcal{L}_{\text{cls}} + \alpha_{\text{pres}}\, g_{p,k} \nabla_{\hat{z}_0} \mathcal{L}_{\text{pres}}\big)$, $\alpha_{\text{cls}} = 4.0$, $\alpha_{\text{pres}} = 0.05$; gradients through the VAE decoder $D$, one backward pass per loss |
| Trade-off in Eq. 1 | classification vs. preservation, no single fixed weight: set per pass by $1.15\, \alpha_{\text{cls}}\, g_{c,k}$ against $\alpha_{\text{pres}}\, g_{p,k}$ (above), and in the reward by $\lambda_{\text{fid}}$ and $\lambda^R_{\text{pres}}$ (Table R1) |
| Guidance ramp | $w(i) = \text{clip}\big((i - \lfloor nr \rfloor)/\max(1, n - \lfloor nr \rfloor),\ 0,\ 1\big)$ over the $n$ kept steps $i = 0, \ldots, n-1$; no gradient is taken where $w(i) = 0$; its maximum, $(n - 1 - \lfloor nr \rfloor)/(n - \lfloor nr \rfloor)$, is below 1 |
| Lung mask $S$ | TorchXRayVision PSPNet (`chestx_det`) on the image resized to $512^2$ and rescaled to $[-1024, 1024]$; sigmoid, maximum over the left- and right-lung channels, resized bilinearly to $256^2$ |
| Object features $O$ | YOLOv8n (`yolov8n.pt`) layers 0–9, global average pooled, frozen, on the $256^2$ image replicated to 3 channels |
| Edit mask $m$ | $S(x_a)$ resized bilinearly to $32^2$; dilated by a $5 \times 5$ stride-1 max-pool (2 latent cells, $\approx 16$ px); feathered by a separable Gaussian, $\sigma = 1$ latent cell, radius 3, zero padding; clamped to $[0, 1]$, not thresholded |
| Background lock | after every step, $z \leftarrow m \odot z + (1 - m) \odot \tilde{z}_a$, where $\tilde{z}_a$ is the latent of $x_a$ forward-noised to the next timestep with fresh noise, and used clean after the last step |
| Grad-CAM arm only | $m \leftarrow m \cdot \big(0.15 + 0.85\, \widetilde{\text{CAM}}^{\,0.75}\big)$; CAM at layer `features`, normalized by its 0.99 quantile at $32^2$, Gaussian blur $\sigma = 0.8$. The trained Grad-CAM arm uses `features.denseblock3` |

The 0.15 cut-off is a heuristic operating point. A ratio of 0.5 is only an illustration, meaning validators reproduce half the guidance shift, and is not a threshold.

**Bootstrap intervals for $\rho_v$.** 10,000 resamples, with replacement, of the $N = 200$ image indices of a group. Numerator and denominator are recomputed on the same re-

Table R5. Settings that differ between code versions. "Default runs" is the value in the tables above.

| Setting | Default runs | Other code versions |
|---|---|---|
| Guidance classifier $f$ | `rsna+chex` mean | `rsna` only (single-classifier arm) |
| Step budget $T$ | 25 | 8 |
| Episode end | validity or $t = T$ | $t = T$ only |
| $L_1$ budget $B$ in validity | $\infty$ | 0.02 |
| Policy trunk | MLP | convolutional |
| LoRA rank / targets | 8 / attention | 16 / + `ff.net.0.proj`, `ff.net.2`, `proj_in`, `proj_out` |
| LoRA epochs | 30 | 15 |
| $a_1$ mapping | $0.30 + 0.90\,\sigma(a_1)$ | $0.50 + 2.50\,\sigma(a_1)$ |
| $R_{\text{fid}}$, $\lambda_{\text{fid}}$ | mean $L_1$, 2.0 | MSE, 0.5 |
| $\lambda^R_{\text{pres}}$ | 3.0 | 0.5 |
| $\lambda_{\text{hit}}$, $\lambda_{\text{stop}}$ | 2.0, 2.0 | 6.0, 8.0 |
| Lock / preservation anchor | episode's original image | previous pass's output |
| Saturation penalty | 0.01 | absent |
| 24-image probe holdout | removed from training pool | absent |

sampled indices, and the 95% interval is taken from the 2.5th and 97.5th percentiles.

**Lesion Hit Rate.** Defined for p→n only, on images with an RSNA opacity annotation. For image $i$ with annotated region $A_i$, hit rate $h_i = \sum_{p\in A_i}\delta_p / \sum_p \delta_p$, null $\pi_i = |A_i|/(HW)$, lift $= h/\pi$. $A_i$ is the union of the image's boxes, rescaled from the native DICOM size to the $256^2$ grid and rounded outward to whole pixels. Reported lift is the mean of the per-image ratios $h_i/\pi_i$, not a ratio of pooled sums; images with $\pi_i \in \{0, 1\}$ are skipped. The reported $t$ and $p$ for lift against chance are from a paired $t$-test of $h_i$ against $\pi_i$. Paired arm comparisons use the images both arms share: paired $t$-test and Wilcoxon signed-rank test on per-image lift differences, $d =$ mean/standard deviation (ddof 1) of those differences, and a 10,000-resample percentile bootstrap CI of the mean difference.

# E. Additional Details

## E1. Controls the comparison requires

A control-arm comparison is only as trustworthy as the plumbing beneath it, and three preconditions we had assumed did not hold during development. (i) A single dataset object was shared by LoRA fine-tuning, the PPO pool and the metrics scan, so evaluation images had been seen during fine-tuning. We replaced it with one deterministic, persisted, *patient-level* split shared by every stage, computed once because generation runs as independent array processes that cannot recompute an RNG-based split consistently. It assigns 80% of patients to training and 20% to test (seed 0; one image per patient in RSNA, so 21,347 training and 5,337 test images) and is stored with a fingerprint of the dataset. In every reported arm, generator fine-tuning, PPO rollouts and the validity probe use only the training fold, and every evaluated image is in the test fold. (ii) Training was unseeded: two identical submissions produced two different policies, which is survivable for a headline number but not for a comparison in which arms must differ by treatment alone. Every reported run now uses split seed 0, LoRA seed 0 and PPO seeds 1 (n→p) and 2 (p→n); evaluation uses base seed 0, with per-image seed equal to the base plus the image's global index ($+10^6$ for p→n). Seeding fixes the random streams but not floating-point nondeterminism, so repeated runs are not guaranteed to be bit-identical. (iii) The control arms had never been exercised. Wiring them up surfaced a routing bug under which every reduce silently fell back to the trained arm's shards, which would have made "the policy contributes nothing" an artefact of a missing flag, and a shared output directory under which each arm overwrote the previous one's metrics.

A CPU-only invariant suite pins the LHR calibration, split disjointness and three further properties whose silent failure would yield a plausible-looking wrong number rather than a crash. Every training and evaluation run writes a manifest giving its code revision and configuration, and every evaluation shard carries a configuration stamp from which the transfer-ratio denominator is read (main paper Sec. 4.3). We report these because each was, at the time it was found, invisible in the outputs: all three would have produced a well-formed table.

## E2. Clinical proxy metrics

We report two clinically motivated proxies under the default configuration: the Cardio-Thoracic Ratio (CTR), the ratio of cardiac to thoracic width from the heart and lung segmentation masks, and a costophrenic-recess score (SCP) approximating pleural blunting, each tested with a paired $t$-

test on the within-image delta (real minus counterfactual).

CTR shifts by $+0.0127$ ($t = 3.882$, $p = 1.4 \times 10^{-4}$) in n→p and by $+0.0047$ ($t = 0.478$, $p = 0.634$) in p→n: a small shift, significant in n→p only, with an absolute magnitude ($\leq 0.013$) far below the metric's clinical resolution (its diagnostic threshold is 0.5). This is consistent with the intended behaviour of the background lock, since the cardiac silhouette lies largely outside the lung mask that gates editing; we read it as a shift too small to act on either way, not as a clinically directional CTR effect. SCP's own computation flags itself as degenerate under this configuration (relative spread 0.19% across the four image groups), so its delta and $t$-test are withheld rather than reported on what is, numerically, a near-constant signal. Since it is built on the feature norm of a COCO-pretrained detector rather than a dedicated recess detector, we do not consider it informative in its current form. Neither proxy supports a directional claim about pleural or cardiac status.

## E3. Comparison to prior methods

We do not compare head-to-head with prior counterfactual generators, for two reasons. First, published values are not comparable to ours: they are computed with other classifiers, feature extractors and evaluation groups, and none of ECED, FiRe or DocVCE reports a lesion-anchored metric or the held-out target shift. Second, running them under our protocol was not feasible for this submission: DiME, ACE and FastDiME guide a pixel-space diffusion model trained on the target dataset, and FiRe a Pixel Mean Flow model, none of which exists for RSNA at our resolution and split; training one per method is a separate project. We therefore report no literature reference values, and we scope every claim about the gap between CV and the anchored metrics to our system.

The closest thing to a baseline we can run under an identical protocol is the generator's own default schedule without a policy: one classifier-guided, background-locked latent diffusion pass with hand-set strength and guidance (the "no policy" row of Table E2). It shows the same pattern as the trained policy: n→p CV 0.940 against a 3-validator mean transfer of 0.307 and `nih` 0.101. So the gap is not specific to the RL scheduler, though it may still be specific to classifier-guided latent editing with our guidance models.

## E4. Lock-width sensitivity (exploratory)

The width of the background lock's editable band (main paper Sec. 3.2: dilation and Gaussian feathering of the lung mask, both in latent cells) was set once and not tuned. As an exploratory check we regenerated the first 50 images of each direction with the default policies under three settings: d0 (dilation 0, no feathering), d2 (dilation 2, feathering 1.0; the default setting, so a rerun) and d4 (dilation 4, feathering 2.0). LHR uses the 47 annotated p→n images. Table E1 gives each setting's paired difference from the d2 rerun, with 95% paired percentile-bootstrap CIs (10,000 resamples of the images, drawn jointly across settings; transfer as a ratio of means; CV also with an exact McNemar test). As the rerun-noise reference we use the difference between the d2 rerun and the earlier evaluation of the default on the same images, and count a difference as exceeding rerun noise when its CI lies entirely outside $\pm$ that reference. The reference itself has CIs excluding zero on several metrics, so a CI that excludes zero is not on its own evidence of a setting effect. Every contrast here is across GPU types, and for p→n metrics that reference evaluation used the n→p adapter for both directions, so its band also contains that change.

Narrowing the lock (d0) raises edit concentration ($+0.028$ n→p, $+0.035$ p→n) and lowers the fraction of pixels changed ($-0.060$, $-0.065$); it also moves two n→p validator ratios beyond noise, in opposite directions (`nih` $-0.059$, `pc` $+0.019$). Widening it (d4) lowers LHR lift by 0.130 (95% CI $[0.038, 0.247]$) and makes edits slightly more diffuse (concentration $-0.007$ and $-0.018$, fraction changed $+0.018$ and $+0.048$). The other differences, including CV and mean transfer in both directions, are within rerun noise; d0's p→n CV drop of 0.140 (1 vs. 8 discordant images, $p = 0.039$) has a CI that excludes zero but not the noise band. With 50 images and a single rerun pair as the noise reference, we read these as indications of sensitivity, not as a tuning result.

## E5. Same-image runs: default, no policy, fixed schedules, second seed

Table E2 compares five runs on the same 400 test images with the same per-image seeds: the default policy, the generator's default schedule without a policy, a fixed schedule that holds every control at the policy's mean action over its evaluation steps (same step budget and validity-based stopping, no training), a tuned fixed schedule chosen without RL, and a second PPO seed of the default configuration. Each direction uses its own adapter, except the second seed, whose two directions share one; both of its directions were resumed from checkpoints after preemption (n→p at update 15, p→n at update 80) without saved random-number state. The fixed schedule's raw actions are, for n→p, $(-9.114, -0.671, -2.007, 2.446, 3.041, -6.357, 4.074, 6.550, -1.090)$ (strength 0.200, three passes, guidance bias 1.299), and for p→n, $(1.705, 1.700, 0.432, 0.486, -1.607, -0.260, -1.142, 2.968, -0.608)$, the means over 201 and 1,093 policy steps. The tuned fixed schedule is selected on training-fold images only, by the quantity PPO maximizes, the mean discounted episode reward. Candidates are 54 constant schedules on a coarse grid (strength $\{0.25, 0.45, 0.65\}$, classifier scale $\{0.50, 0.85, 1.15\}$, guidance bias $\{0.85, 1.05, 1.27\}$, two or three passes; every

Table E1. Lock-width sensitivity, first 50 images per direction (LHR: 47 annotated). d2 is the default setting, rerun. Differences are setting minus d2 rerun, with 95% paired bootstrap CIs. Rerun noise is d2 rerun minus the earlier default evaluation on the same images. †CI entirely outside ± rerun noise. ‡CI excludes zero but not the rerun-noise band.

| Metric | d2 (rerun) | Rerun noise | d0 − d2 | d4 − d2 |
|---|---|---|---|---|
| CV, n→p | 0.960 | 0.000 [0.000, 0.000] | −0.020 [−0.060, 0.000] | 0.000 [0.000, 0.000] |
| CV, p→n | 0.740 | −0.060 [−0.140, 0.000] | −0.140 [−0.260, −0.040]‡ | +0.040 [−0.040, +0.120] |
| LHR lift (p→n) | 2.489 | −0.034 [−0.114, +0.043] | +0.013 [−0.096, +0.124] | −0.130 [−0.247, −0.038]† |
| Edit conc., n→p | 0.248 | 0.000 [−0.002, +0.001] | +0.028 [+0.025, +0.032]† | −0.007 [−0.010, −0.004]† |
| Edit conc., p→n | 0.317 | −0.005 [−0.013, +0.002] | +0.035 [+0.025, +0.043]† | −0.018 [−0.025, −0.010]† |
| Frac. changed, n→p | 0.664 | 0.000 [0.000, +0.001] | −0.060 [−0.072, −0.050]† | +0.018 [+0.008, +0.025]† |
| Frac. changed, p→n | 0.629 | +0.001 [−0.008, +0.009] | −0.065 [−0.074, −0.056]† | +0.048 [+0.037, +0.060]† |
| Transfer mean, n→p | 0.262 | −0.005 [−0.015, +0.002] | −0.016 [−0.032, −0.001]‡ | +0.004 [−0.007, +0.015] |
| Transfer mean, p→n | 0.251 | −0.038 [−0.062, −0.014] | −0.025 [−0.052, +0.003] | +0.046 [+0.015, +0.077]‡ |
| nih, n→p | 0.042 | −0.012 [−0.040, +0.010] | −0.059 [−0.101, −0.018]† | +0.017 [−0.010, +0.047] |
| pc, n→p | 0.442 | −0.002 [−0.007, +0.004] | +0.019 [+0.006, +0.031]† | −0.002 [−0.018, +0.012] |
| mimic_ch, n→p | 0.304 | −0.002 [−0.006, +0.002] | −0.008 [−0.024, +0.007] | −0.004 [−0.013, +0.006] |
| nih, p→n | 0.124 | −0.060 [−0.105, −0.021] | −0.038 [−0.082, +0.007] | +0.073 [+0.018, +0.129]‡ |
| pc, p→n | 0.320 | −0.004 [−0.040, +0.030] | −0.029 [−0.070, +0.010] | +0.033 [+0.001, +0.068]‡ |
| mimic_ch, p→n | 0.307 | −0.049 [−0.097, −0.006] | −0.009 [−0.061, +0.045] | +0.031 [−0.008, +0.070] |

other control at the midpoint of its range) and a local grid around the policy's mean action (three levels each of strength, classifier and preservation scale, strength boost and, for n→p, ramp start: 243 schedules for n→p, 81 for p→n). Each is scored on 12 training-fold images per direction, and the best of each grid on 36 more; the stop bonus that the coarse grid's neutral stop threshold forgoes is added back before ranking. Both selected schedules come from the coarse grid: strength 0.25, classifier scale 1.15, guidance bias 0.85 and two passes for n→p, and strength 0.25, classifier scale 1.15, guidance bias 1.27 and three passes for p→n.

Against the default policy (paired, LHR on the 148 annotated p→n images, CV with exact McNemar tests): the fixed schedule differs by $-0.071$ in LHR lift ($[-0.161, +0.019]$, $p = 0.12$, $d = -0.13$), 0.000 in n→p CV (0 discordant) and $-0.010$ in p→n CV (14 vs. 16 discordant, $p = 0.86$); the second seed by $-0.117$ ($[-0.227, -0.006]$, $p = 0.039$), $-0.030$ (0 vs. 6, $p = 0.031$) and $-0.175$ (7 vs. 42, $p = 3.6 \times 10^{-7}$); no policy by $-0.532$ ($[-0.645, -0.419]$, $p = 1.9 \times 10^{-16}$), $-0.005$ (3 vs. 4, $p = 1.0$) and $-0.445$ (1 vs. 90, $p = 7.4 \times 10^{-26}$). Relative to no policy, the fixed schedule recovers 87% $[71, 102]$% of the policy's LHR-lift gain and 98% $[86, 111]$% of its p→n CV gain (paired bootstrap over images, 10,000 resamples); in n→p the policy has no gain to recover. The tuned fixed schedule differs from the default policy by $-0.187$ in LHR lift ($[-0.279, -0.095]$, $p = 1.0 \times 10^{-4}$, $d = -0.33$), $-0.005$ in n→p CV (1 vs. 2 discordant, $p = 1.0$) and $+0.050$ in p→n CV (22 vs. 12, $p = 0.12$); from the mean-action schedule by $-0.116$ in LHR lift ($[-0.217, -0.013]$, $p = 0.025$) and $+0.060$ in p→n CV (21 vs. 9, $p = 0.043$); and from no policy by $+0.344$ in LHR lift ($[+0.260, +0.432]$, $p = 1.6 \times 10^{-12}$). Its mean p→n transfer is below that of the policy and of the mean-action schedule (paired differences $-0.065$ $[-0.085, -0.045]$ and $-0.032$ $[-0.052, -0.013]$); in n→p the three schedules differ by at most 0.005. A search over fixed schedules by the training reward thus matches the policy's validity but not its lesion targeting or transfer, which the reward does not measure.

## E6. Edit locality and FID uncertainty

Table E3 is computed offline from the saved counterfactuals of the first three runs of Table E2 (no regeneration). With $\delta = |x_c - x|$ on the $256^2$ grid: the share of $\sum \delta$ inside the editable region ($m > 0.5$, where $m$ is the background lock's soft mask of main-paper Sec. 3.2, recomputed from the source image and upsampled bilinearly) and inside the undilated lung mask ($S > 0.5$), each beside its area fraction, which is what a spatially uniform edit would score; the fraction of pixels with $\delta > 0.01$, the same fraction measured against the VAE round trip $D(E(x))$ (posterior mean), and the fraction the round trip alone changes; and the share of $|x_c - D(E(x))|$ outside $m > 0.5$. FID uses the main paper's features, with a 95% interval from 500 bootstrap resamples of both sets; KID uses the same features (cubic polynomial kernel, unbiased, 100 subsets of 100). The recomputed FIDs match the evaluation's to within 0.01.

Three observations. (i) The lock works as intended at the level it acts on: relative to the VAE round trip, only 3–7% of the change lies outside the editable region. Most of the difference outside it, and about half of all pixels changed by more than 0.01, is VAE reconstruction error that any latent-space editor with this autoencoder incurs, so the main paper's "fraction of pixels changed" overstates the generative edit; against $D(E(x))$ it is 0.41–0.44 for the default policy. (ii) The edits are lung-localized but diffuse within the lung:

Table E2. Five runs on the same 400 test images and seeds. CV with Wilson 95% CIs; LHR with a $t$-interval (concentration and fraction changed have $t$-interval half-widths $\leq 0.01$); transfer per validator with a 10,000-resample bootstrap, the mean with a paired bootstrap over per-image deltas. †Second seed: shared adapter, resumed training.

| Dir. | Run | CV | LHR lift | nih | pc | mimic_ch | Transfer mean | Edit conc. | Frac. changed |
|---|---|---|---|---|---|---|---|---|---|
| n→p | default policy | 0.945 [0.904, 0.969] | — | 0.044 [0.005, 0.082] | 0.465 [0.436, 0.494] | 0.333 [0.302, 0.364] | 0.281 [0.259, 0.303] | 0.254 | 0.647 |
| | no policy | 0.940 [0.898, 0.965] | — | 0.101 [0.059, 0.142] | 0.474 [0.444, 0.503] | 0.345 [0.314, 0.376] | 0.307 [0.284, 0.329] | 0.268 | 0.641 |
| | fixed schedule | 0.945 [0.904, 0.969] | — | 0.044 [0.004, 0.083] | 0.469 [0.439, 0.499] | 0.334 [0.302, 0.365] | 0.282 [0.260, 0.304] | 0.254 | 0.647 |
| | tuned fixed schedule | 0.940 [0.898, 0.965] | — | 0.039 [0.002, 0.077] | 0.466 [0.437, 0.494] | 0.326 [0.294, 0.358] | 0.277 [0.256, 0.299] | 0.252 | 0.637 |
| | second seed† | 0.915 [0.868, 0.946] | — | 0.204 [0.161, 0.246] | 0.458 [0.428, 0.488] | 0.334 [0.304, 0.365] | 0.332 [0.310, 0.355] | 0.279 | 0.675 |
| p→n | default policy | 0.670 [0.602, 0.731] | 2.595 [2.450, 2.741] | 0.165 [0.138, 0.193] | 0.306 [0.278, 0.335] | 0.357 [0.317, 0.397] | 0.276 [0.254, 0.298] | 0.314 | 0.634 |
| | no policy | 0.225 [0.173, 0.288] | 2.064 [1.961, 2.166] | 0.113 [0.083, 0.143] | 0.245 [0.218, 0.275] | 0.149 [0.099, 0.197] | 0.169 [0.146, 0.192] | 0.278 | 0.599 |
| | fixed schedule | 0.660 [0.592, 0.722] | 2.525 [2.385, 2.664] | 0.151 [0.124, 0.179] | 0.261 [0.236, 0.288] | 0.319 [0.278, 0.357] | 0.244 [0.224, 0.264] | 0.312 | 0.631 |
| | tuned fixed schedule | 0.720 [0.654, 0.778] | 2.408 [2.303, 2.514] | 0.180 [0.151, 0.210] | 0.205 [0.180, 0.230] | 0.249 [0.208, 0.290] | 0.211 [0.190, 0.232] | 0.305 | 0.637 |
| | second seed† | 0.495 [0.426, 0.564] | 2.479 [2.350, 2.608] | 0.154 [0.126, 0.182] | 0.218 [0.192, 0.246] | 0.252 [0.208, 0.295] | 0.208 [0.187, 0.230] | 0.298 | 0.620 |

Table E3. Edit locality and distribution metrics, $n = 200$ per direction and run. "in $m$" and "in lung": share of $\sum |x_c - x|$ inside the editable region and the lung mask, with the area fraction (uniform-edit null) in parentheses. "changed": fraction of pixels with $|\cdot| > 0.01$ against $x$, against $D(E(x))$, and for $D(E(x))$ itself. "out of $m$": share of $\sum |x_c - D(E(x))|$ outside the editable region. Locality values have 95% $t$-interval half-widths $\leq 0.017$. KID $\times 10^3$, mean $\pm$ s.d. over subsets.

| Dir. | Run | in $m$ (null) | in lung (null) | changed vs. $x$ | changed vs. $D(E(x))$ | $D(E(x))$ floor | out of $m$ vs. $D(E(x))$ | FID [95% CI] | KID$\times 10^3$ |
|---|---|---|---|---|---|---|---|---|---|
| n→p | default policy | 0.835 (0.590) | 0.653 (0.339) | 0.647 | 0.436 | 0.499 | 0.036 | 22.39 [20.43, 24.56] | 69.3 ± 3.6 |
| | no policy | 0.845 (0.590) | 0.675 (0.339) | 0.641 | 0.429 | 0.499 | 0.031 | 15.92 [14.33, 17.78] | 48.4 ± 3.1 |
| | fixed schedule | 0.836 (0.590) | 0.655 (0.339) | 0.647 | 0.434 | 0.499 | 0.034 | 22.51 [20.78, 24.49] | 70.0 ± 3.2 |
| p→n | default policy | 0.808 (0.495) | 0.663 (0.268) | 0.634 | 0.406 | 0.474 | 0.054 | 4.76 [3.86, 5.87] | 12.5 ± 1.4 |
| | no policy | 0.734 (0.495) | 0.558 (0.268) | 0.599 | 0.359 | 0.474 | 0.068 | 2.69 [2.31, 3.24] | 5.1 ± 0.5 |
| | fixed schedule | 0.798 (0.495) | 0.647 (0.268) | 0.631 | 0.402 | 0.474 | 0.058 | 4.40 [3.61, 5.27] | 11.6 ± 1.4 |

about 82% of the mass sits in a region covering 49–59% of the image, while edit concentration stays at 0.25–0.31. In p→n the policy places more of the change in the lung than the no-policy schedule (0.663 vs. 0.558, null 0.268), consistent with its LHR gain, and the fixed schedule is close to the policy (0.647). (iii) FID and KID favor the no-policy schedule in both directions, with non-overlapping bootstrap intervals, while the policy and the fixed schedule are indistinguishable. Since the feature extractor also guides generation (main-paper Sec. 4.1), we read this only as evidence that the stronger schedule moves counterfactuals further in the guidance classifier's feature space, not as a realism ranking.